

Ankit Kumar Ojha

AI Engineer — RAG — Agents — MCP — Python — LangChain — FastAPI

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Professional Summary

AI Engineer (B.Tech Data Science, 2025) building and shipping RAG systems and AI agents in Python using LangChain, LangGraph, FastAPI, Postgres and MCP. Live projects include VoxServe voice + text support copilot (59/60 tool accuracy), QueryPilot safe text-to-SQL (5/5 eval), DriftGuard AI (28/30 auto-fixed, re-runs down 60%), and Veritas AI with page-level citations and evals (Hit@5, MRR, NDCG). Focused on practical AI that works in production.

Projects

VoxServe — Voice + Text Support Copilot — Live Oct 2026

GitHub: <https://github.com/ankitojha15/VoxServe> | Live: <https://voxserve.onrender.com/>

- Tech Stack: Python, FastAPI, LangGraph, Postgres + pgvector, Redis, FastMCP, Groq, LiveKit, Langfuse, Docker, Render.
- Built support agent that checks live orders, creates tickets, and answers policy questions with [file page X] citations; retrieval is intent-routed over separate policy / ticket pgvector collections.
- Routed all data access through MCP tools (get_order_status, get_policy, create_ticket, search_past_tickets) with API-key auth and audit log; LangGraph agent escalates to human below 0.7 confidence, 2 missed voice turns auto-create ticket.
- Validated on 60 queries (50 golden + 10 adversarial): 59/60 tool accuracy, 59/60 faithfulness, 7/7 unsafe blocked; p95 0.13s text, 1.1s voice first-audio, \$0.002/query, eval-gated CI.

QueryPilot — Natural Language to SQL Agent — Live Sep 2026

GitHub: <https://github.com/ankitojha15/QueryPilot> | Live: <https://querypilot-ra2d.onrender.com/>

- Tech Stack: Python, FastAPI, LangGraph, LangChain-Groq, Postgres (Supabase), Redis (Upstash), sqlglot, Docker, Render.
- Built text-to-SQL agent that turns plain English into safe SELECT-only queries using Groq with schema.yml rule-book; unclear questions ask back (city/state, 7/30/90 days), sensitive ones need approval.
- Enforced guardrails with sqlglot: SELECT-only, server-forced org_id from login token, PII email blocked, LIMIT 200 auto-added, audit logged; destructive queries auto-blocked.
- Added per-org Redis cache (300s), FastAPI APIs (/login, /clarify, /query, /audit) with real rows; 5/5 eval pass, live with demo logins across 2 orgs.

DriftGuard AI — Self-Healing Data Pipeline — Live Sep 2026

GitHub: <https://github.com/ankitojha15/DriftGuard-AI> | Live: <https://driftgaurd-ai-night-guard.streamlit.app/>

- Tech Stack: LangGraph, LangChain, Groq, Postgres (Neon), FastAPI, Streamlit, Slack, Render.
- Detect schema drift by diffing Postgres information_schema between prod and sandbox copies.
- Draft fix SQL with Groq (LangChain), test on sandbox with rollback on fail.
- Apply to prod only after human approval in Streamlit (FastAPI /detect, /approve).
- Live with daily cron and Slack alerts; 28/30 golden breaks auto-fixed, re-runs down 60%.

Veritas AI — Hallucination-Guarded RAG over PDFs — Live Sep 2026

GitHub: <https://github.com/ankitojha15/Veritas-AI> | Live: <https://knowledgebaseai-pzi7.streamlit.app/>

- Tech Stack: Python, LangChain, FAISS, Hugging Face Embeddings, Groq, FastAPI, Streamlit.
- Built LangChain RAG system answering PDF questions with page-level citations using Hugging Face embeddings, parent-child chunking, hybrid dense + BM25 retrieval with RRF and cross-encoder reranking.
- Used Groq with query rewriting and citation verification; built FastAPI + Streamlit chat UI with streaming.
- Built 60-query eval harness (50 normal + 10 adversarial) reporting Hit@5, MRR, NDCG; tested end-to-end.

Technical Skills

- **Languages:** Python, SQL
- **Data/ML:** Scikit-learn, NumPy, Pandas
- **AI/GenAI:** LangChain, LangGraph, LangSmith, RAG, AI Agents, MCP, Tool Calling, Memory, Observability, Prompt Engineering, LLM Evals
- **Backend:** FastAPI (REST, Async), SQLAlchemy, Pydantic
- **Data/AI Infra:** FAISS, BGE / HuggingFace Embeddings, Hybrid Retrieval (BM25, RRF, Reranking), Groq, OpenAI, Gemini APIs, Postgres (Neon, pgvector, Supabase), MySQL, Redis, sqlglot
- **Tools:** Git, GitHub, Docker, Render, Langfuse, LiveKit

Education

Oriental Institute of Science & Technology, Bhopal
B.Tech - Data Science

Sep 2021 – Jun 2025